

Hands-on Exercise No. 1
DigiSkills 3.0 Batch-03
AI Using Python

Total Marks: 10
Due Date: 30/04/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
 - It is submitted after due date.
 - It is not in the required format (.doc or .docx)
 - It does not open, or file is corrupt.
 - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- Apply **conditional statements in Python** to solve real-world problems such as tax calculation based on different conditions.
- Work with **lists and dictionaries** to transform data and map values (e.g., creating key-value pairs from a list).
- Understand and implement **basic Object-Oriented Programming (OOP)** concepts by designing classes and building a simple chat system with multiple functionalities.

Problem Statement

In this hands-on-exercise, you are required to apply your understanding of **basic Python programming, data structures, and Object-Oriented Programming (OOP)** to solve practical problems. You will write programs using conditional statements for calculations, work with lists and dictionaries to transform data, and design a simple chat system using classes.

The objective is to strengthen your problem-solving skills and understand how different programming concepts are applied together to build real-world applications.

You can use any programming environment such as **Jupyter Notebook, Google Colab, VS Code, or any other Python editor** to write and execute your programs.

After completing each of the following given task, take screenshots of your code and its output, and paste them into a single MS Word (.doc/.docx) file as your final solution submission.

Tasks:

1. Write a program that takes salary as input. Using conditional statements, calculate the final tax based on the following rules: **(Marks 3)**
 - If the salary is less than 30,000 → Tax rate is 5%
 - If the salary is between 30,000 and 70,000 → Tax rate is 15%
 - If the salary is greater than 70,000 → Tax rate is 25%
2. Given a list of words: words = ["apple", "banana", "kiwi", "cherry", "mango"] **(Marks 3)**
Create a dictionary that maps each word to its corresponding length.
Example Output: ({"apple": 5, "banana": 6, "kiwi": 4, "cherry": 6, "mango": 5})
3. Create a Chat System using Object-Oriented Programming (OOP) concepts. **(Marks 4)**
You need to create the following classes:
 - User
 - Message
 - ChatRoom

The system should implement the following functionalities:

- Sending messages
- Viewing chat history
- User joining and leaving the chat room

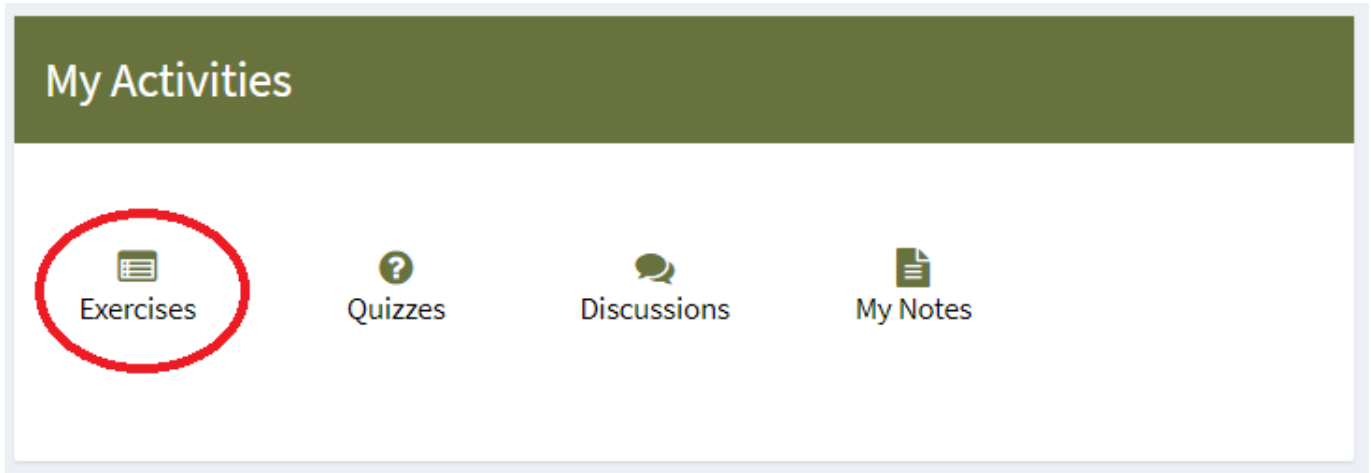
NOTE: Save the code file at your end and only provide Screenshots of each step in MS Word File.

BEST OF LUCK ☺

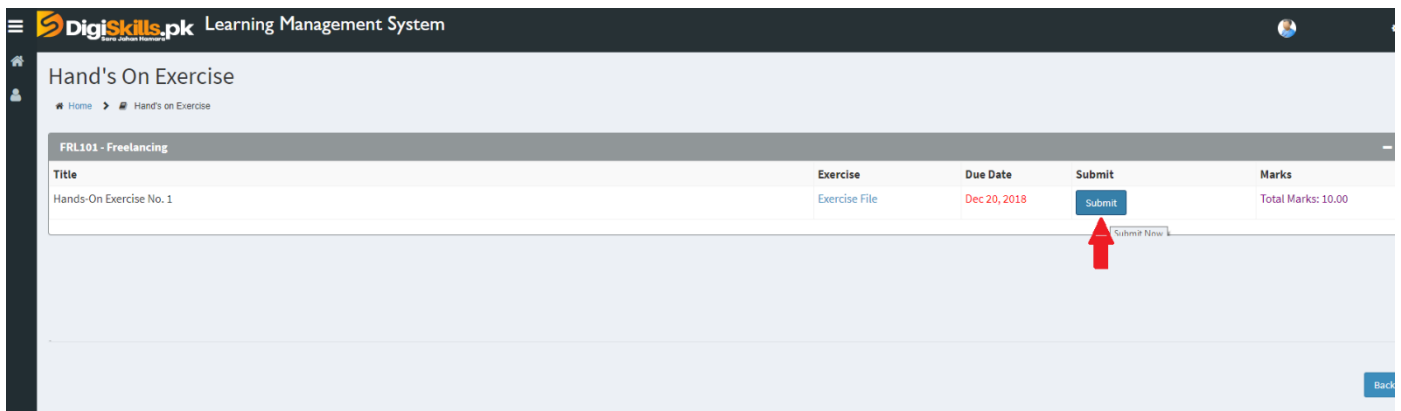
How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format